

Repsol Bio Electra Synth

Synthetic ester dielectric oil — Inhibited

PRODUCT DATA SHEET

Description

BIO ELECTRA SYNTH is a dielectric oil for transformers based on synthetic esters, manufactured from raw materials especially selected to obtain a high-performance fluid. It offers excellent oxidation stability whilst also having a very low pour point, making it suitable for cold climates. It provides increased safety for installations and the environment as a result of its excellent fire properties and is highly biodegradable. It is therefore suitable for cases where a fluid with a high flash point is required or when priority is given to sustainability and environmental factors.

Classification and standards

BIO ELECTRA SYNTH offers high oxidative stability, is highly biodegradable, covers a wide range of working temperatures, and has excellent fire properties (fire point over 300 °C, classified as a K3 type fluid in accordance with IEC 61100). It complies with the IEC 61099 standard.

Technical specifications

Values according to IEC 61099 standard.

Properties	Units	Method	Min.	Max.	Typical value
Physical properties					
Appearance		Visual		Bright and clear	
Colour		ISO 2211		200 Hazen	30
Density at 20 °C	g/cm ³	ISO 12185		1.000	0.969
Viscosity at -20 °C	mm ² /s	ASTM D445		3000	1196
Viscosity at 40 °C	mm ² /s	ASTM D445		35	27.4
Pour point	°C			-45	-50
Chemical properties					
Acidity	mg KOH/g	ISO 6618		0.03	0.01
Water content	mg/kg	IEC 60814		200	22
Electrical properties					
Dielectric breakdown voltage	kV	IEC 60156	45		77
Dielectric dissipation factor at 90 °C, 50 Hz		IEC 60247		0.03	0.005
Resistance at 90 °C (continuous current)	G Ohm.m	IEC 60247	2		7.4
Oxidation stability — IEC 61125C (164 h)					
Soluble acidity	mg KOH/g	IEC 61125C			0.04
Volatile acidity	mg KOH/g	IEC 61125C			0.01
Total acidity	mg KOH/g	IEC 61125C		0.3	0.05
Sludge totals	% (m/m)	IEC 61125C		0.01	0.01

Properties	Units	Method	Min.	Max.	Typical value
Health, safety and the environment					
Fire point	°C	ISO 2592	300		308
Flash point	°C	ISO 2719	250		255
Biodegradability after 28 days	%	OECD 301B	60		72
Aquatic ecotoxicity	mg/l	OECD 201/202/203	100		>1000

Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.